Bangladesh University of Professionals (BUP)

Faculty of Science & Technology

Department of Information & Communication Technology (ICT)

B.Sc. in Information & Communication Engineering

Course Title: Data Communication & Computer Networks

Course Code: ICE- 2207

Term Paper on

Network Technology in Industrial Automation

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Semester: July-December, 2022

Date of submission: 21-03-2023

Acknowledgement:

First and foremost, we want to thank and honor God, Almighty Allah, for providing us the willpower and blessings necessary to successfully finish our term paper.

We would want to thank our course instructor, Dr. S.M. Salim Reza, in the most sincere manner possible. The BUP assistant professor who offered us the chance to complete our term paper. We have been inspired to work on our paper with all of our might under his wise guidance. Our deepest gratitude to him for leading us at this time with his invaluable supervision.

We would especially want to thank our friends who helped and encouraged us throughout the paper-writing process.

We also want to express our sincere gratitude to everyone who helped us finish our term paper, whether directly or indirectly.

Abstract:

This paper discusses the advent and development of networking technologies and industrial automation, specifically the use of networking technologies in industrial automation. The history, methodology and current state of networking technologies, particularly those used by companies to achieve automation, have been covered in this paper. Industrial automation so far has been mainly achieved by successfully integrating three networking technologies: Ethernet, wireless communication, and the Internet of Things (IoT) in industrial practices. Alongside the pros and cons of industrial automation, career prospects in this field both in the Bangladesh market and the global market have also been covered in this paper.

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Chapter – 01

Introduction

1.1 Network Technology:

The numerous devices and methods needed to build, maintain, and secure computer networks are together referred to as network technology. A computer network is a group of connected devices, including computers, servers, routers, switches, and other hardware, that enables them to interact and share resources. This technology has become incredibly important in an age of digital transformation, allowing companies to expand their reach and employee capabilities, while providing them with a faster and more reliable way of accomplishing tasks than ever before.

Network technologies come in a wide variety, including:

Local Area Networks (LANs): A LAN is a network that encircles a single building, workplace, or other small space. Ethernet cables or Wi-Fi are frequently used to connect LANs together.

Wide Area Networks (WANs): A wide geographic area, such as a city, state, or country, is covered by a wide area network (WAN). Typically, leased lines, satellites, or the internet are used to connect WANs.

Wireless Network: Wireless networks is a kind of network which allow devices to connect to a network without the use of physical connections. Some examples of wireless network is Wi-Fi, Bluetooth etc.

Network Security: Technologies for securing networks against unauthorized access, attacks, and other security risks include firewalls, intrusion detection systems, and other devices.

1.2 Industrial Automation:

Automation of manufacturing operations and other industrial activities is referred to as industrial automation. It involves using tools like robots, machines, sensors, and other equipment to carry out tasks that were traditionally done by people or call for a high degree of accuracy or consistency.

Manufacturing, agriculture, and transportation are just a few of the industries that use industrial automation. By decreasing the possibility of human error, it can assist increase productivity, lower labor costs, and improve product quality. Some examples of industrial automation are robotics, programming logic controllers, machine vision systems etc.

1.3 Network used in Industrial Automation:

Digital technologies are used in industrial automation to automate numerous production-related tasks. It involves integrating different equipment and systems so that they can interact and function as a single, integrated system. Industrial companies are now able to manage their

production processes more effectively, boost production yields, and reduce the need for manual labor. The foundation of this automation is networking technology, which offers a productive way to link sensors, actuators, and other control systems.

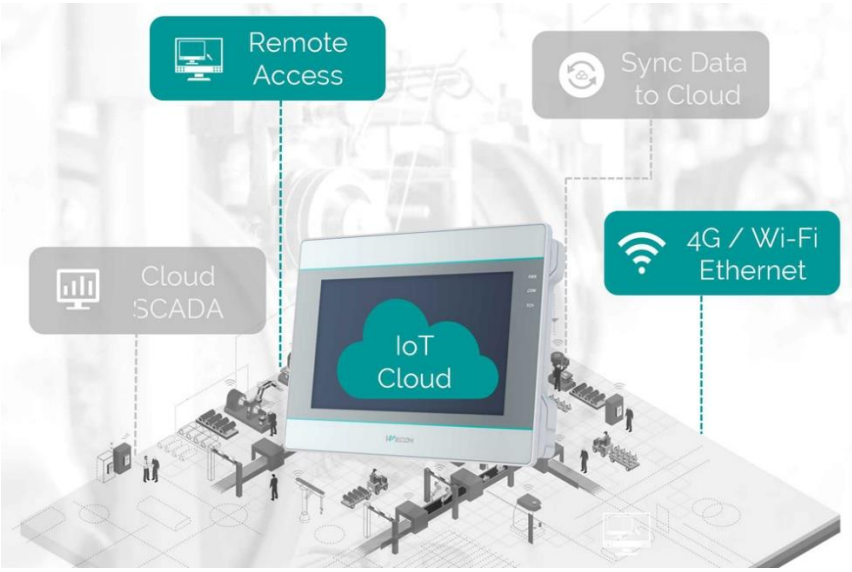

Fig-1: Network technology used in industrial automation

Chapter – 02

History of Networking Technology Used in Industrial Automation

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2.1 History of Networking Technology:

Today, a variety of networks and protocols operate to enable data communication, making networking technology widely used around the globe. However, how was this technology created? We'll look at the growth of networking technology from its early stages through the creation of the modern internet in order to gain a better understanding.

The first advancement in networking technology dates all the way back to 1844 when Samuel Morse developed his famous telegraph system. Operated through a series of dots and dashes known as Morse code, this system enabled the transmission of messages over large distances without the need for wires or cables. Things gradually began to accelerate from here. In 1889, Alexander Graham Bell developed the 'Photophone', using acoustic transmission of voice signals over a beam of light. The introduction of the telephone in 1877 only improved the quality of data

communication across networks. The telephone eventually replaced the telegraph as the number one means of data transmission, as it enabled the transmission of audio signals with superior sound quality over larger distances. In the 1970s, advances in networking technology led to the development of the Internet Protocol Suite (IP). The 2000s were a time of modernization for networking technology. The development of Wi-Fi network technology and wireless routers revolutionized the way people connected to the internet.

2.2 History of Industrial Automation:

Industrial automation refers to the use of computer-controlled machines for automated processes, such as manufacturing and materials handling. Automation can be used to improve efficiency and safety in the workplace. Industrial automation has been around since the start of the industrial revolution and has evolved over the years.

Industrial automation first emerged from the foundations of the industrial revolution. In the early 1900s, electrical systems began supplanting mechanical linkages in the application of industrial automation. This period saw a number of important inventions – most notably the relay and the programmable logic controller (PLC). The first PLC was developed in the 1940s and laid the groundwork for further electrical automation technology. The 1960s were a major period for the advancement of industrial automation, as microprocessors and computers began to be used in the control of automated machines. Today, industrial automation is used in a variety of industries, from large-scale manufacturing to retail and service businesses. Automation is used in warehousing and distribution to optimize delivery times, in the food industry to increase safety and efficiency, and in the automotive industry to reduce costs by streamlining production processes.

2.3 History of Networking Technology Used in Industrial Automation:

Industrial automation has been around since the dawn of the Industrial Revolution in the early 1800s and automation technology has developed and grown with each new iteration.

Automation technology has evolved to include more advanced elements, such as computerized systems, sensors, and robotics. The earliest applications of automation technology in industry date back to the late 19th century when gears, motors, and mechanical relays were used to control and power certain processes, such as pumps and valves. However, these early systems were limited in scope and were only able to automate very basic functions. With the advancement of technology, the 20th century saw automation technology become more sophisticated with the use of computers and dedicated networks that allowed for the communication of data between machines and allowed them to be automated with far more complexity. During this time, industrial networks, such as the widely popular PLCs (Programmable Logic Controllers), were developed and widely adopted. The advent of the internet and the World Wide Web in the late 20th century allowed for the instantaneous communication of data over vast distances. The next step in the evolution of

industrial automation was the emergence of the Internet of Things (IoT). IoT is a network of physical devices, such as sensors and actuators, that allow machines to communicate with one another and with their operators. The IoT allowed manufacturers to remotely monitor, control and optimize the performance of their industrial automation systems.

The continuous development of networking technology used in industrial automation has helped to introduce significant increases in production efficiency and cost savings, while at the same time providing improved safety conditions in the workplace.

Chapter – 03

Technology Used in Industrial Automation

3.1 Ethernet:

Ethernet is a type of technology used for communication between computers and other devices over a local area network, often referred to as LAN. It is a particularly resilient and reliable type of technology that can function across a long distance, where the distances are typically less than 100 meters.

At its basic level, Ethernet consists of a network of interconnected components which transmit data to each other, often referred to as nodes. It uses a protocol called the Carrier Sense Multiple Access (CSMA) protocol, to control the flow of data between the nodes. This ensures that there is no wastage of available resources and that all nodes have a fair chance of accessing the network at any given time.

One of the biggest benefits of Ethernet technology is its low price and high reliability. This is partly why it has become so popular in the modern world. It is relatively easy to set up and maintain, making it well suited to both home and business use. Ethernet is also capable of transmitting data at very high speeds and can process data at the speed of Gigabits per second.

There are potential drawbacks to using Ethernet technology however. One of the primary issues is its limited range. While it is capable of long-distance transmissions, this range is typically limited to 100 meters. Another issue is that whilst Ethernet is capable of very fast speeds, this is only when the available bandwidth is not overloaded. If many users are connected to a single network, then the speeds can be significantly reduced, as the available bandwidth will be divided amongst them.

Overall, Ethernet is a very reliable, cost-effective and secure form of networking technology. It is popular in both home and business environments thanks to its flexibility and scalability. However, its limitations in terms of range and the potential for decreased speeds when many

users are connected mean that many times faster, more modern solutions such as fiber-optic cables are preferable for larger networks.

3.2 Wireless Communication:

Wireless communication is the transmission of information from one point to another without the use of physical wires or cables. In this way, it eliminates the need for physical connections or even lines of sight. Wireless communication networks are becoming increasingly popular due to their affordability, convenience and portability. Wireless communication allows users to move freely without the need for fixed wiring, and can provide access to a wider range of services

There are various types of antennas used in wireless communication, such as directional antennas, omnidirectional antennas and beam antennas. These antennas are designed to send out a focused wave signal, enabling more efficient and reliable data transmission.

A wireless network, a type of computer network that use radio waves rather than physical cabling, is also built using wireless communication. The key advantages of wireless networks are their scalability and flexibility, which allow for easy expansion and modification as user needs change. Since wireless communication does not require any cabling or wires, wireless networks also lower installation costs.

The main drawback of wireless communication is the possibility of electromagnetic radiation and physical objects interfering with data transmission across airways. Also, because wireless communication networks don't need physical access to the networking hardware in order to transmit data, they are susceptible to hacking. Users must therefore make sure that their wireless networks are secure.

Wireless communication is undoubtedly becoming a common kind of communication in a variety of applications, despite its flaws. Wireless communication is playing a more and bigger role in our environment, from mobile phones and public Wi-Fi networks to military and governmental operations. So, in order to ensure that data carried across wireless networks is safe and secure, it is crucial to design strong encryption algorithms and use smarter network management strategies.

3.3 Internet of Things (IoT):

The Internet of Things, commonly referred to as IoT, is one of the most revolutionary technologies in the world today, able to connect physical devices to the Internet and integrate them into the digital landscape. By connecting our everyday objects to the Internet, an immense protentional for growth, interaction, and innovation is unlocked for new industries, organizations, and individuals.

Essentially, the IoT consists of physical objects that are connected to the Internet, either through a wireless connection or hard wire. These objects can be anything from a computer, a smartphone, a television set, a smart speaker, an internet-connected refrigerator, a home security system, a smartwatch, a fitness tracker, and more. These objects can then be equipped with sensors that

measure and collect data about our environment, for example, temperature, humidity, and light sensors. The data that is collected is then sent to the Cloud where it is processed and monitored to help us better understand our environment.

The Internet of Things is set to revolutionize the way we live, work, and play. Through its potential applications, business owners have access to inexpensive and reliable remote systems that can automate fundamental tasks, such as gate security or employee time-tracking. Additionally, the IoT can be used to increase interdisciplinary collaboration in research, healthcare, and education. For example, doctors and nurses may be able to access and monitor vital signs of their patients in real time, or students may be able to access course materials from anywhere. The implication of the IoT for our daily lives is extremely expansive, as it will allow for more efficient, automated, and secure processes that weren't possible before.

The biggest challenge to the successful implementation of the Internet of Things is the security and privacy of our data. Since this technology relies on the transmission of sensitive data across the network, it is incredibly vulnerable to malicious actors who could gain access to our networks and our information. This presents a significant risk to our personal data, as well as our businesses. To ensure the safe and secure use of the IoT, organizations must implement adequate security protocols and employ a zero-trust model when it comes to granting access to their networks.

The extent of the Internet of Things' reach is just starting to be felt, but improvements in its development and security will undoubtedly continue over time. Although the potential of the IoT can be daunting, incidents of data breaches are on the decline, and the integration of the technology into our everyday lives is becoming easier and more accessible, allowing us to get a better quality of life. By continuing to refine and improve the way we use the Internet of Things, our lives can become safer, healthier, and better connected.

Chapter – 04

Ethernet

4.1 History

Ethernet is a network communications staple, allowing computers in a local area network (LAN) to connect and share resources with one another. It is the most common type of connection found in homes, businesses, and schools today. To understand the current version of Ethernet, it is important to look back to its history.

It was in the 1970s that Ethernet began to develop. It was a collaborative effort led by Robert M. Metcalfe, a professor at the Massachusetts Institute of Technology. Metcalfe was researching local networks and came up with a concept that he called "Alto Ethernet". The prototype he created was

a system of several interconnected Alto personal computers that were able to communicate using the original Ethernet protocol. In the 1980s, 3Mb Ethernet was replaced by another version of the protocol. This time the new version of Ethernet was 10 Mb/s, and was officially adopted in 1983 as IEEE 802.3. Years later, IEEE 802.3u was released which provided faster speeds of 100 Mb/s.

By the end of the 1990s, the need for more robust speeds was evident. This led to the release of IEEE 802.3ab, which provided ultra-fast gigabit speeds of up to 1000 Mb/s. Later on, IEEE 802.3an was released in 2006 providing even faster speeds of 10Gbps.

Ethernet has been adopted as the industry norm for networking in recent years. It connects PCs, printers, scanners, routers, and other LAN-connected devices. Ethernet's continued extensive use in residences, businesses, and educational institutions is ensured by its ease of use, speed, reliability, and backward compatibility. It has undergone a lengthy, complex evolution, yet up to this point, it has continued to meet technological demands.

Ethernet still reigns supreme over local area networks, despite recent advancements in alternative technologies like Wi-Fi. It has provided a trustworthy and efficient platform for communication and resource sharing since its start many years ago, which accounts for its success.

4.2 Methodology:

Ethernet is a networking technology that is used by local area networks (LANs) in computing. The linked devices can access a number of services via a common communication channel. As a result, Ethernet is extensively used in workplaces, educational institutions, residential settings, and many other places.

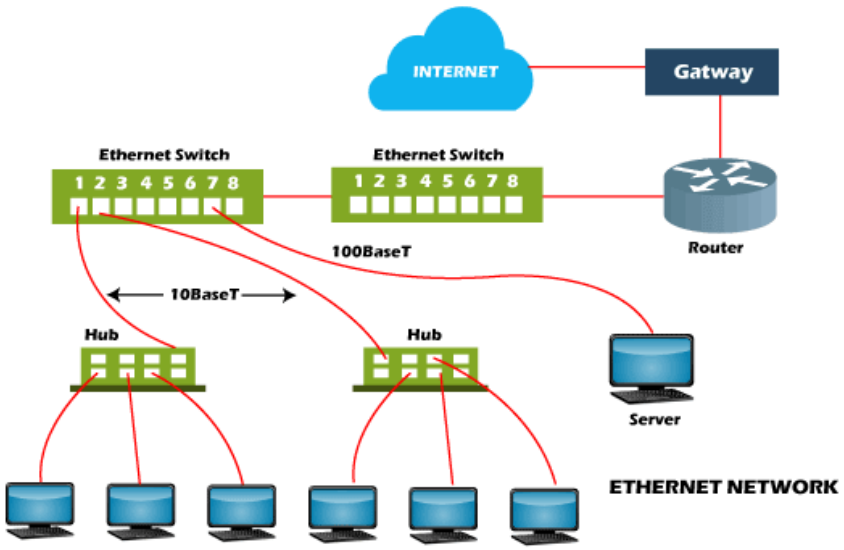

Fig-05: Ethernet.

The Institute of Electrical and Electronics Engineers (IEEE) Ethernet standard contains guidelines on Ethernet (IEEE 802.3) which illustrates Ethernet as a multiaccess bus topology which uses a single cable to transport data over a series of time slots. The time slots with fixed widths are known as frames.

Ethernet supports a variety of media at the physical layer, including coaxial cables, twisted pair cables, and fiber optic connections. These physical media are configured for various purposes where a variety of signals are needed to be transmitted. Coaxial cables, for instance, are appropriate for locations with low noise, whereas fiber optic cables are utilized for greater distances.

At the link layer, Ethernet relies on carrier sense multiple access/collision (CSMA/CD) to enable multiple devices to coexist peacefully on the same medium. This method evenly distributes the bandwidth among all devices on the network and enables transmission of multiple frames at once.

Apart from the physical and link layers, Ethernet also has a protocol layer. This layer is responsible for ensuring that all frames sent over the network have their information correctly formatted and that the integrity of the frames is intact. The IEEE 802.3 standard dictates the use of the Ethernet Protocol (EP) and the Internet Protocol (IP) to provide these features.

Ethernet has enabled high-speed, reliable data transmission over LANs, which in turn has led to the development of many modern technologies. It is widely used in situations where many users need to access the same services simultaneously, such as office networks and online gaming.

The methodology of Ethernet is important for understanding how it works, and also for understanding the various adaptations it has undergone over the years. Ethernet has come a long way and continues to be the go-to choose for many network-based applications. Its implementation is an example of how a network protocol can be standardized and adapted to the changing needs of users.

4.3 Status-quo:

Ethernet is the most widely used network technology in the world. It is a protocol that allows computers and other electronic devices to access and exchange information using a common communications standard. It is a medium for transferring data from one point to another, usually through a wired network. Ethernet has been in used for about thirty years and serves as the basis for the world wide web, the backbone of the internet, and a large part of most local area networks. Ethernet has evolved from the original copper wiring to wireless connections and is now finding its way into automotive systems, industrial automation, and consumer applications.

The traditional Ethernet protocol is based on the CSMA/CD (Carrier Sense Multiple Access with Collision Detection) standard. In this system, two or more nodes simultaneously use the same data link to share information. Each of the nodes monitors the medium for incoming data, and then waits its turn to transmit. If two nodes try to transmit at the same time, the collision is detected, and the nodes are forced to back off and retry later.

As Ethernet has grown in popularity and its technology has become more widely accepted, the standards governing its usage have grown as well. The IEEE 802.3 standards are the most important standards for Ethernet. It is comprised of many different sub-standards, the most commonly used being 802.3U for 100Mbps Fast Ethernet and 802.3ab for Gigabit Ethernet.

For Ethernet networks to function properly, they must include appropriate networking devices and Ethernet cabling. Networking devices can consist of Ethernet switches, routers, bridges, and access points. These devices act as the control layer between the client computers and the network. The cabling used to link these devices to each other is typically either twisted-pair cables or optical fiber cables. All modern Ethernet networks use Category 5e (Cat 5e) or higher twisted-pair cables for cable runs greater than 9 meters and Category 6 (Cat 6) or better for greater than 15 meters.

Ethernet cabling can also be used in the office or home. Home users can benefit from 10/100Mbps Ethernet network for connecting their computers to the internet and for file and printer sharing. For enterprise networks, a Gigabit Ethernet switch is recommended for data speeds over 100Mbps

The current status quo of Ethernet allows it to be used in a wide variety of applications. Because of its widespread acceptance, Ethernet is the technology infrastructure behind the Internet, enabling global connectivity and communication. Ethernet is also used in corporate networks to connect computer systems, servers, and other peripheral devices. Now, as cars, homes, and industrial systems are all becoming more connected, Ethernet technology is being adopted in those systems as well.

Ethernet technology is expected to become even more ubiquitous in the years to come. There is an ongoing expansion of the products and technologies that will use Ethernet and the number of applications for which Ethernet is the ideal solution. The various waves of technological advancement, from cloud computing to IoT, will bring further advances in Ethernet technology and usage. The continual evolution of Ethernet will help ensure that it remains the de facto networking protocol of choice for many years to come.

Chapter – 05

Wireless Communication

5.1 History:

Wireless communication has come a long way from its inception in the late 19th century. Long-distance communication technology has evolved into a sophisticated and essential technology that permeates all aspect of our everyday lives. With the advent of wireless routers, cell phones, Wi-

Fi, and Bluetooth, the ability to communicate without the use of cables has become crucial in today's technologically advanced society. Guglielmo Marconi's transatlantic transmission of the first radio signal in 1897 marked the beginning of wireless communication. The advent of radio communication was signaled by this. Radio was a common form of communication by the early 1900s, and it was utilized by the military, ships at sea, and even casual hobbyists. Radio engineering made advancements and adjustments during the ensuing decades. Smaller and more potent radios and transceivers were developed, and the range of signals they could transmit increased.

During the 1950s, watching television broadcasts had become a regular form of entertainment. In the 1960s and 1970s, cordless telephone technology was developed and quickly used. As a result, telephones might be used without having to physically hook a cord into a network. The technology was improved during the ensuing several decades and became a commonly utilized method of communication in households all over the world.

Cellular networks first appeared in the 1980s and 1990s. Telephone communication between two portable devices using radio frequencies was made possible by cellular technology. The telecommunications sector was subsequently revolutionized by this technology, ushering in the era of internet-based communication.

The invention of Wi-Fi in the late 1990s greatly increased the speed and range of wireless connections. Wi-Fi allows for fast, reliable connections over large areas and is now used in homes, offices, and public spaces all around the world. In addition to Wi-Fi, Bluetooth technology emerged in the early 2000s. This allows devices to connect to each other without the need for cables or distance restrictions.

Wireless communication technology has advanced greatly since its inception in the late 19th century, and it will do so in the 21st century as well. Wireless communication technology has continuously improved to meet our demands since the first transatlantic radio wave through the Internet of Things, allowing us to stay in touch in ways that were before impractical.

5.2 Methodology:

The way businesses, organizations, and people communicate with one another has been changed by wireless technology. Today, we can communicate and receive information quickly and easily via a variety of channels, including emails and text messages. The process for wireless communication is developing along with technology.

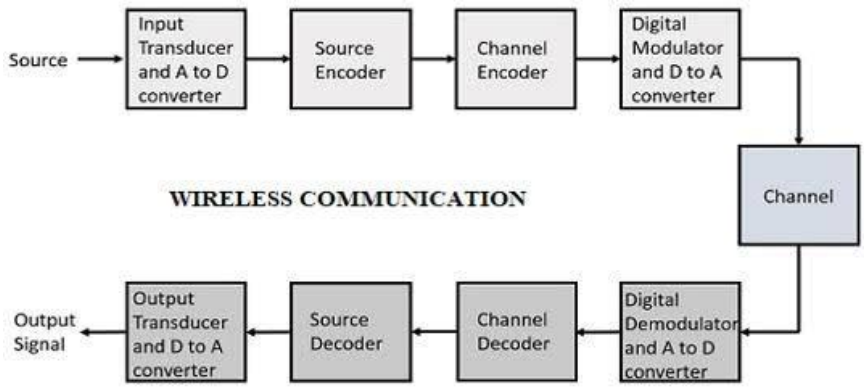

Fig-03: Block diagram of Wireless communication.

Let's examine the various wireless communication techniques and their applications.

Cellular Technology

The most popular kind of wireless communication is cellular technology. Radio waves are used by cell phones to connect to cellular service providers. Generally, when people use their cell phone to make a call, it transmits encoded data to the adjacent cell tower (this is known as the uplink). The tower gathers the information and sends it to the right person (this is known as the downlink). A cellular device has to have a working battery, an antenna that can communicate with the tower, and an active service plan for this to occur.

Cellular technology is used in a wide range of applications, such as mobile communication, GPS systems, and even military applications. In addition to voice calls, cellular technology can also be used to transmit data and photos. This technology is also becoming increasingly important for the Internet of Things (IoT), as it provides a good range, reliable connection and low power consumption.

Satellite-based Communication

Satellite-based communication is a technology that uses satellites to relay and receive information from Earth. This method is commonly used for broadcasting and two-way communication (for example, mobile networks). One big advantage of using satellite-based communication is that the satellite can be used to expand the coverage area and reach further than any terrestrial medium.

Satellite-based communication systems are used for applications that require a weak signal, such as communications in rural or remote areas. They are also used by maritime vessels, aircraft, and other modes of transportation. It is also an excellent solution for providing global access to the internet.

Wi-Fi

Wi-Fi technology is the most popular method of wireless communication today. Wi-Fi is based on the IEEE 802.11 family of standards, which allows wireless devices to communicate with each other by using radio waves. Wi-Fi is used for numerous applications, including internet access, gaming consoles, media streaming, and phone calls.

Wi-Fi can be used with a range of different devices, such as laptops, smartphones, game consoles and smart TVs. With Wi-Fi, users can access the internet, share data, and transfer files quickly and conveniently. Wi-Fi also supports high-speed wireless data transfer, so it is an ideal solution for streaming video and other large files.

Bluetooth

A short-range wireless protocol called Bluetooth is used to link two or more devices together. The Bluetooth protocol enables users to communicate data, connect peripherals, and exchange files between two compatible devices by transmitting and receiving encoded data. It is mostly used to connect entry-level devices including speakers, headsets, mice, keyboards, and mice. Streaming media, including audio, video, and graphics, is another application for Bluetooth technology.

5.3 Status-quo:

Wireless communication has become increasingly more prevalent since the 1990's. This technology has become a staple of everyday life and its use is now taken for granted. From cellphones to tablets, computers and GPS systems, wireless technology has shaped the way we connect with each other, access information, and conduct our work.

Wireless communication has become even more essential with the rise of the internet of things, or IoT, which connects everyday devices from clocks to home appliances. This has created new applications and services, allowing people to monitor, control and automate various functions for comfort, security and efficiency.

Wireless communication, however, does come with its challenges. Both traditional wired and wireless networks suffer from interference caused by distance, environmental factors, or other networks congesting the airwaves. Without strong signal strengths and wide coverage, wireless communication can become overwhelmed and unreliable. Another challenge is security – wireless networks require encryption protocols in order to prevent hacking, as well as authentication procedures to allow certain users access.

Fortunately, recent advances in wireless technology have enabled faster data speeds and more reliable communication. 4G and 5G cellular networks now transmit data at much faster rates, enabling new applications that require high bandwidth communication. Additionally, new technologies such as millimeter wave and massive MIMO have been deployed in order to extend the reach of wireless networks.

Bluetooth and NFC (Near Field Communication) are two emerging technologies that enable short-range connections, commonly used for hands-free device control, or for contactless payments.

Progress in this area is allowing these standards to be used for more complex tasks, such as transferring large amounts of data or creating networks of connected devices.

Finally, satellite communications are becoming increasingly popular for use in remote locations where terrestrial connections are impractical. These systems operate independently of traditional cellular and wireless networks, but can also use them for ancillary services. While traditionally used for broadcasting television and radio signals, satellite communication is becoming increasingly accessible for data transmissions and internet access.

Overall, wireless communication continues to grow and its impact can be seen in almost every aspect of life. With new technologies, faster speeds and longer reach, wireless communication has expanded the possibilities of the internet of things and provided instructions to streamline everyday life. With further research and innovation, wireless communication will continue to shape the way we communicate.

Chapter – 06

Internet of Things (IoT)

6.1 History:

Internet of things (IoT) is the concept of connecting physical objects to the internet in order to communicate and exchange data. IoT has a long and rich history that dates back to the 1980s, when major technology companies began to explore the feasibility of networks connecting everyday objects. Over the years, numerous inventions and discoveries have helped to shape the modern IoT landscape.

The history of IoT begins with the invention of the Radio Frequency Identification (RFID) technology in the mid-1980s. RFID is a wireless technology used for communication and data exchange between two or more objects. RFID is the foundation for many of the modern applications of IoT, including smart locks and security systems, electronic payments, and inventory tracking. RFID tags and readers have been used to monitor livestock, as well as to wearables and medical devices.

In 1998, Kevin Ashton, a British technologist, developed the term “Internet of Things.” His vision of the future was that everyday objects around us could be connected to the internet and be able to identify and interact with each other. He proposed that computers would be embedded into everyday objects and act as wireless sensor networks, allowing them to interact and exchange data.

In the early 2000s, corporate and government institutions began to explore the potential of connected devices, and more companies began to emerge in this space. Major technology players such as Microsoft and Cisco began to invest in IoT research and development. In the 2010s, the

term IoT began to gain traction in the media and technology community and the concept started to become a reality.

Today, IoT technology is used in all sorts of applications, including smart cities, industrial automation, and beyond. IoT devices are becoming commonplace in almost every industry and are expected to grow in number and usefulness over the course of the next decade.

The history of the Internet of Things is a long, ongoing journey. Each step forward in technological exploration brings us closer to the realization of implementing connected devices in our everyday lives. From security to convenience, the potential of IoT is vast, and it's only a matter of time before the world around us becomes more connected, intelligent, and secure. In the near future, the applications of IoT will be virtually limitless.

6.2 Methodology:

The methodology of IoT can be broken down into four distinct stages: design, development, deployment, and maintenance. Each of these stages requires a meticulous and thorough approach in order to ensure the best possible outcome.

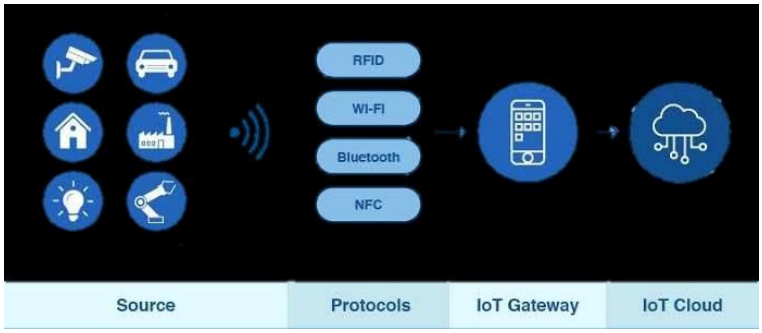

Fig-04: Internet of Things (IoT).

Understanding the principle of each step helps to understand the methodology as a whole.

Design:

Any IoT project's success depends on the design phase, which comprises choosing the project's goals and objectives. Also, it entails examining the needs of the network, the sensors, and any additional components that might be required. The architecture is also defined during the design phase, along with any unique features or performance considerations. Finally, this stage entails assessing potential hazards connected to the project and thinking about the security consequences.

Development:

The development phase starts when a design plan has been established. The actual coding and hardware development start at this point. This often demands the most resources and requires knowledge of both hardware and software engineering. To make sure the finished result fulfills

the stated requirements, the project's numerous components are now polished, tested, and certified. To ensure that the system functions as intended, it is crucial to test it in the environment in which it was built.

Deployment:

Once the development phase is complete, the deployment phase begins. This phase involves making sure the network is connected, integrating the components into the system, and setting up any additional software or hardware. It also requires configuring the various components to work together in a cohesive manner. Furthermore, during this phase, the security of the overall system should be verified to protect against potential threats.

Maintenance:

Once the deployment phase is complete, the system must be kept up to date and functional. Regular maintenance is essential in order to identify any issues early and address them as needed. This should include regular security updates as well as regular testing and performance evaluations to ensure that the system continues to function correctly.

IoT approach is, in general, a step-by-step procedure that necessitates commitment and meticulous attention to detail. To achieve the intended final result, every part of the project should be carefully taken into account. To guarantee the greatest results, the four stages of design, development, deployment, and maintenance must be carried out expertly. Over the duration of the procedure, the system must be dependable and secure to prevent any potential threats related to IoT.

6.3 Status-quo:

IoT has the potential to revolutionize the way people interact with the physical world through interconnected sensors, machines, and objects. It has the potential to revolutionize the way businesses monitor and control their physical environment, and the ways in which we access data, provide services, and monitor the health of our environment.

While the potential of IoT is immense, so too are the challenges. In 2017, IoT was estimated to be worth $7.1 trillion, with the global market reaching $19.9 trillion by the end of 2024.

Currently, within the consumer space, the largest applications of IoT technology is seen in smart home products, such as smart thermostats, connected appliances, surveillance cameras, and automated lighting systems. IoT is also making major impact in terms of transportation, specifically connected cars. As vehicles become more integrated with the internet, they can offer drivers a significant range of benefits, mostly in terms of convenience and safety.

In the industrial space, the IoT market is estimated to grow to $1.1 trillion by 2022. A range of industrial applications are utilizing IoT technology, such as predictive maintenance, asset monitoring and management, connected logistics, and cloud-driven analytics. IoT has enabled

businesses to monitor their physical environments more effectively, allowing for increased efficiency and cost savings, as well as better operational agility.

The Industrial Internet of Things (IIoT) has been making major inroads in other areas, such as agriculture, healthcare and finance. In terms of agriculture, IoT connected devices allow for the collection and analysis of data regarding climate, soil content, pest infestations and more. This data can be used to optimize crop yields, improve the efficiency of irrigation systems and reduce costs associated with crop diseases.

However, while the potential of IoT is significant, a number of key issues are holding back the widespread adoption of the technology. Primarily, there is an issue surrounding security and privacy. By connecting various physical objects and data points to the internet, the attack surface of a business is increased dramatically. Insecure networks, connected devices and inadequate authentication mechanism can make businesses susceptible to cyber-attacks. Furthermore, the data that is collected and transferred between devices must be secure and stored appropriately, as well as being managed in compliance with data protection regulations.

Additionally, there is an issue of interoperability within the current IoT landscape. In a fully merged IoT network, devices should be able to communicate seamlessly across manufacturers, locations and platforms. While many manufacturers are striving towards this goal, the reality is that it is far from being a reality.

Overall, the Internet of Things is in a promising state right now, but there are urgent problems that must be solved before it can be widely used. Together with the difficulties of interoperability and data consumption, security and privacy must be addressed. As a result, there is a lot of potential for the technology to disrupt established sectors even more and open up new opportunities. IoT can genuinely integrate into the operations of any business and create a host of new opportunities with the proper solutions.

Chapter – 07

Industrial Automation in Bangladesh Market

In the last few decades, Bangladesh has seen an increasing trend in industrial automation. This has been largely due to the growing demand for efficient and cost-effective automation solutions in order to sustain the country's rapid production growth. With the rapid advances in industrial automation technologies, state-of-the-art automation solutions are now widely accessible to all businesses, even in the smaller and medium-scale sector. This has opened up new market opportunities for Bangladesh's industrial automation industry.

Industrial automation has become a necessity for most businesses in Bangladesh to remain competitive in the global market. By improving process efficiency, reducing costs, and streamlining operations, automation technology has allowed businesses to greatly increase production capacities and reduce labor costs. Automation technology has also enabled manufacturers to reduce scrap materials, improve accuracy and quality, and improve delivery time. In the long run, industrial automation allows businesses to not only increase their productivity but also reduce their operating costs and increase their profits.

In Bangladesh, a wide range of automation products is available for different purposes, from simple sensors and controllers to sophisticated robots and automated machines. Motors, sensors, controllers, actuators and communication modules are the key components of industrial automation systems. The availability of advanced automation products on the market has revolutionized the way Bangladeshi businesses operate. Automation systems have allowed businesses to automate almost every operation, from inventory and product tracking to production and maintenance processes. Automated factories, warehouses and distribution centers have vastly improved operational efficiency and reduced operational costs significantly in Bangladesh.

However, the rise of automation technology in Bangladesh has also faced some challenges. For example, with the spread of automation technology and the related rapid technological advancement, the skills gap between engineering and IT professionals has widened and stranded many unemployed engineer graduates and professionals. Furthermore, some of the most advanced automation solutions, such as robots, are far too expensive for most Bangladeshi businesses to afford. In addition, due to the lack of a proper infrastructure in the country, the implementation of automation solutions can be very challenging.

In order to overcome these challenges, the Bangladeshi Government has invested significantly in developing infrastructure and bringing in high-tech solutions. The Government has established high-tech incubation centers, invested in artificial intelligence research and development, and provided subsidies to promote the use of automation solutions among small and medium-scale businesses.

In addition, the Government has launched initiatives such as “Make in Bangladesh” and industrial automation clusters. This has allowed companies to avail training and resources to encourage the use of automation solutions. The Government has also offered tax discounts and incentives to further encourage the automation industry. This, in turn, has enabled companies to access cutting-edge automation solutions and reduce operational costs.

Furthermore, industry collaborations, innovation networks and support from academic institutions have allowed businesses to better deploy automation solutions and increase their productivity.

In conclusion, the introduction of industrial automation has drastically enhanced the operational efficiency of businesses in Bangladesh. Automation solutions have allowed businesses to reduce operational costs, increase production capacity, improve accuracy and quality, and enhance customer service. The Government of Bangladesh has also taken several steps to further promote the use of automation solutions in the country. Through public-private partnerships, industry collaborations and a supportive Government, Bangladesh has emerged as a thriving hub for industrial automation.

Chapter – 08

Industrial Automation in International Market

Industrial automation is rapidly transforming the international market by revolutionizing the way industrial products and materials are produced and distributed. It is the use of technology and computers to control and monitor production processes that were previously performed manually. Automation is allowing for improved accuracy, higher efficiency, and increased safety across a wide range of industries. As automation technologies have become more available and cost-effective, they have been increasingly adopted in the international market.

The introduction of industrial automation has begun to reshape the production and distribution of industrial materials and products. Automation has enabled industries to streamline processes, reduce labor costs, and increase accuracy. Additionally, automation allows for the production of products with reduced lead times and improved quality. Automation has also enabled manufacturers to respond faster to changing market conditions and customer needs. As a result, the production process is more efficient and economical, while products are more likely to satisfy customer requirements.

In addition to improved efficiency and reduced costs, automation in the international market has also opened up new and exciting opportunities. Automation systems allow for the analysis of large amounts of data from various sources to assess production, identify trends, and make improvements. This capability has enabled industries to develop more effective strategies for managing their operations and navigating the global market.

The most significant benefit of industrial automation in the international market is the ability for industries to produce and distribute products at a faster rate than ever before. Automation has greatly reduced the need for manual labor, allowing for higher throughput and reduced costs. This has enabled manufacturers to quickly respond to changes in the global market, while producing products with higher accuracy and quality.

Automation has also enabled manufacturers to produce products and materials with lower environmental impacts. Automation systems have been designed to reduce the use of resources such as water and electricity, while minimizing waste. Additionally, automation can help reduce the use of chemicals and other pollutants, leading to improved environmental safety standards.

Finally, automation has enabled companies to improve customer satisfaction by providing better and faster customer service. Automation systems have the ability to monitor customer orders, shipping information, and customer feedback. This allows companies to quickly identify customer needs and delivery times, while also responding to customer complaints and dissatisfaction more quickly.

Industrial automation is rapidly transforming the international market, from changing the way industrial materials and products are produced and distributed, to providing exciting opportunities for companies to become more flexible and responsive to customer needs. Automation has enabled increased efficiency and reduced costs, while ensuring higher levels of accuracy and quality. The international market is now more competitive than ever, and companies that embrace industrial automation are well positioned to capitalize on their advantages.

Chapter – 09

Advantages of Industrial Automation

Industrial Automation is the use of automated technology or machines to perform or assist in the production process. Automation in the industrial process has become a necessary tool in helping businesses reduce cost and increase efficiency. The technology has made production operations faster and simpler, as machines can be programmed to do the work instead of relying on manual labor. Here are some of the key advantages of industrial automation.

The first major benefit of industrial automation is cost reduction. Automation reduces the demand for manual work, which lowers labor costs. In addition to being less maintenance-intensive and more effective than manual labor, automated machinery. As a result, businesses are able to significantly lower their operational costs. Industrial automation also boosts productivity by boosting the throughput of the production process, which ultimately results in higher profitability.

Second, safety is another significant benefit of industrial automation. Machines are designed to handle dangerous tasks that may be too risky for manual labor. This ensures that staff is out of harm's way throughout the production process, reducing the chances of workplace accidents. Automation also helps to monitor hazardous areas and detect any issues that could lead to a safety hazard.

Third, automation is often used for quality control. Automated machines can be programmed to check for discrepancies and provide information about the quality of a product as it passes through the production line. This helps reduce production defects, resulting in higher quality products. Companies also benefit from improved customer satisfaction, better customer retention and an overall boost in reputation.

Fourth, another advantage of industrial automation is improved efficiency. Automated processes are faster and more efficient than manual labor and can be incorporated into existing operations with minimal disruption. Furthermore, machines can be programmed to detect problems and make adjustments when necessary, resulting in greater efficiency and lower operating costs.

Fifth, industrial automation provides increased production scalability. Automated processes can be quickly adjusted to accommodate changes in production needs. As businesses grow, machines

can be reprogrammed to meet the increased demand, allowing companies to increase their production volume without having to invest in extra labor.

In conclusion, industrial automation has many advantages, including cost savings, improved safety, quality control, increased efficiency, production scalability and flexibility. As the technology continues to evolve and become more accessible, companies of all sizes are investing in automated solutions to help reduce their operating costs and improve the efficiency and quality of their production processes.

Chapter – 10

Disadvantages of Industrial Automation

Industrial automation holds a great deal of promise for businesses, from cost savings to greater efficiency in production, but it does come with some drawbacks. Automation can be expensive to install and maintain, and the responsibilities for managing automation are often unclear. In addition, some experts believe that automation can lead to job losses and unequal access to job opportunities.

Industrial automation has a potentially high starting cost. Installation of machinery, other devices, and all required software, sensors, and controllers are prerequisites for automation. Programmers, engineers, and technicians are frequently needed to set up and maintain automated systems. This can be a considerable cost, especially for small organizations that do not have the money to invest in automated solutions.

Long-term industrial automation maintenance costs are also continuing expenses. In order for a machine to operate well, it needs to be maintained and upgraded, and doing so might be expensive. A business must also account for the price of educating staff members to utilize and maintain automated technology.

The ambiguity surrounding who is accountable for leading and directing the system is another possible drawback of industrial automation. The decision to switch to automated systems should be taken at the managerial level, but many businesses lack a defined procedure for how the system is to be handled and who should be held accountable in the event of a problem. Because of this, workers could feel unsure of themselves and ill-prepared to deal with any issues that may occur.

Furthermore, given that automated systems are capable of managing a variety of activities more effectively than human workers, some experts worry that automation may result in job losses. Employee layoffs and reduced work possibilities could result from this, resulting in unequal access to job opportunities. It should be noted, however, that the majority of automation technologies do not entirely substitute employees, but rather enhance their work, allowing businesses to focus workers on more complex activities and those that call for a degree of human judgment and creativity.

Last but not least, automated systems may be susceptible to disruption from cyberattacks or other unanticipated occurrences. Because automated systems significantly rely on communication between numerous devices and sensors, manufacturing processes are particularly vulnerable to disturbance.

In conclusion, industrial automation can be a powerful tool for increasing efficiency and cost savings in manufacturing processes, but businesses should carefully consider the potential drawbacks before investing in an automated system. Automation requires significant investment, both in terms of the initial cost of installation and ongoing maintenance. Additionally, businesses should ensure that they have a clear system of management in place and consider the potential disruption to employees and customers.

Chapter – 11

Scope for Employment in Industrial Automation

Industrial automation has revolutionized businesses in recent years, with its adoption increasing rapidly in industries everywhere. The scope of employment in industrial automation is vast, as businesses of all shapes and sizes are utilizing this technology to increase their productivity and reduce labor costs. Automation has opened up a range of opportunities for skilled workers who can design and implement these systems, as well as for those who can maintain them.

The main element of industrial automation is industrial robots. These robots are taught to carry out repetitive or accurate activities, such running powered machinery or putting together goods. A higher level of production can be achieved in a shorter amount of time thanks to the design and implementation of automated industrial systems by high-level engineers with expertise in robotic engineering and programming. A greater number of robotics engineers are required to create, maintain, and upgrade robotic systems as automation technology develops to become more sophisticated and affordable.

Software experts with expertise in the development of industrial automation systems are highly sought-after as automation software gets more complex. The automation systems that allow machines and systems to operate automatically improve worker safety and efficiency. The software engineer designs these systems and oversees their implementation.

Industrial maintenance engineers are also in demand as automation technology progresses, as they are responsible for oversight and maintenance of the automated systems. Maintenance engineers inspect and repair robotic systems, check mechanical and electrical parts, and ensure that the systems are running in top condition. They may also provide advice and recommendations on the best ways to implement and get the most from the automation systems.

Another vacancy created by the scope of automation is that of a robotics technician. These skilled workers are responsible for setting up, programming, and providing troubleshooting for robots. Robotics technicians use their expertise to make sure that the robots are operating safely and correctly, and that the tasks for which they are programmed are accurately fulfilled. Those with mechanical and electrical engineering experience, as well as programming skills, have the best chance of securing a job in this role.

Besides these roles, specialists in mechanical engineering, electrical engineering, and computer engineering can also find employment opportunities in driving and managing the implementation of automation technology. Such professionals help create and design the robots, along with other associated equipment that are used in automated processes. Since the integration of automation technologies into businesses of all sizes and industries, the use of artificial intelligence (AI) is also growing, and specialists with knowledge and experience in this field are also in demand.

Finally, the jobs created by the scope of automation span the entire industry spectrum. From operational managers and technical customer service personnel to post-sale personnel who provide training and customer support, these roles all provide opportunities for those looking to specialize in industrial automation.

In short, the scope of employment in industrial automation is broad and varied, offering numerous opportunities for those seeking careers in this field. As technology advances and automation becomes increasingly pervasive, the demand for skilled professionals in automation and related roles is likely to continue to grow.

Chapter – 12

Conclusion

Network technology has become a crucial element in industrial automation. As the industry continues to evolve, the need for automated systems to manage an increasing number of sophisticated processes continues to grow. Network technology has become a critical asset for managing and controlling industrial processes. By leveraging network technology, industrial automation solutions can be designed to improve efficiency and accuracy while also reducing costs.

Network technology is used in industrial automation to provide control and communication between various components that are used in the manufacturing process. This can include motors, sensors, actuators, controllers, and other components. By using a network-based automation solution, components can be connected and data can be transmitted between them. This connectivity allows for faster and more effective control and communication of the manufacturing process. In addition, the ability to share data across the network facilitates better decision making, as the data can be used to identify anomalies or potential manufacturing issues.

Network technology is also being used in industrial automation to improve the security of automated systems. By utilizing identity access management solutions and encryption technologies, data can be protected from unauthorized access. This helps to safeguard not only the data but also the integrity of the industrial automation process. As a result, network technology plays a pivotal role in creating a secure environment.

Network technology has become a fundamental part of industrial automation. It provides a means of control and communication between components, facilitates secure data transmission, enables predictive maintenance, and is critical for the development of the Industrial Internet of Things. As the technology continues to evolve, it will play an increasingly important role in helping manufacturers gain a better understanding of their operations and drive improvements in the industrial automation process.

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